CASE 03 / PORTFOLIO DEMO | FLUTTER

Flutter Task Manager Clean Architecture

A modular task-management demo with Riverpod state and interchangeable data repositories.

Flutter | Dart | Firebase | Mobile App Development | Android App Development

PROBLEM

A task-management demo should let another engineer explore the application without first provisioning a backend. It should also make backend changes understandable without forcing a rewrite of the interface. This portfolio project addresses those needs through a modular Flutter structure, replaceable repositories, and an in-memory default runtime.

CONTRIBUTION

My public Task Manager repository demonstrates Flutter implementation across sign-in and sign-out, a simple task list, state controllers, repository boundaries, and a reusable Material 3 design system. The project separates pure Dart domain entities and interfaces from data implementations and the Riverpod application layer. UI and bootstrap code consume those abstractions through provider wiring.

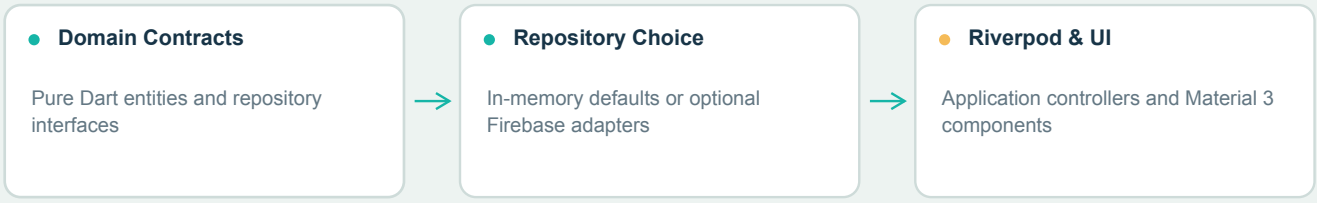
DECISIONS

In-memory authentication and task repositories are selected by default so the demo can be explored without Firebase setup. Runtime flags and provider overrides select the optional Firebase Auth and Firestore implementations, keeping that choice outside the UI. Firebase-backed repositories are created lazily to address initialization timing. Shared tokens, theme builders, buttons, inputs, and cards provide a consistent Android-first Material 3 foundation.

RESULT

The deliverable is a Flutter portfolio app with clear extension points for data access, state, and UI. Its documentation describes tests for models, repositories, controllers, and widgets, alongside a web release build workflow and CI coverage artifacts. These give reviewers specific implementation areas to inspect when discussing maintainability and test organization. The project is presented as an engineering demo with optional backend integrations, rather than a commercial deployment.

ENGINEERING FLOW



EVIDENCE

Flutter Task Manager - public project and README

Material 3 design-system documentation